
Public Spending In New York Towns

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Question:

Does government spending within NY towns come more from local taxes or government aid? (as seen through 3 metrics)

Specifically measuring:

- General Government Spending
- Public Safety
- Social Services

Data

Data Source: Office of the New York State Comptroller, Local Government Bulk Data

- 9981 observations across 914 NY Towns
- 11 years worth of data (2013–2023)

Converted to panel by appending each year into one dataset

gen_gov

Total amount of expenditures for services provided by the governmental entity for the benefit of the public or governmental body as a whole.

Includes:

- Administration
- Zoning and planning
- Operations
- Judgments
- County distribution of sales tax
- Miscellaneous general government

public_safety

Total amount of expenditures for police, fire and other public safety services.

Includes:

- Public safety administration
- Police
- Fire protection
- Emergency response
- Correctional services
- Disaster response
- Homeland security and civil defense
- Miscellaneous public safety

social_services

Total amount of expenditures for providing public assistance for individuals/families economically unable to provide essential needs for themselves.

Includes:

- Social service administration
- Financial assistance
- Medicaid
- Non-Medicaid medical assistance
- Housing assistance, employment services
- Youth services
- Public facilities
- Miscellaneous social services

fed_aid

Total amount of revenues derived from the federal government

Includes:

- Assistance with education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous federal Aid

state_aid

Total amount of revenues derived from State Aid.

Includes:

- Education
- Public safety
- Health
- Transportation
- Social services
- Economic development
- Culture and recreation
- Community services
- General government
- Utilities
- Sanitation
- Miscellaneous State Aid

local_taxes

Total amount of tax revenue within town.

Includes:

- Property Tax
- Non-Property Tax (city income tax)
- Other property Tax (gains from sale of tax-acquired property)
- Sales & Use Tax (Sales Tax)

Clustering Standard Errors

Looks at each individual entity's standard error instead of lumping them all together.

- Not clustering underestimates standard errors

Coefficients don't change when clustering, only standard errors

Clustering Standard Errors

Without Clustered SE:

gen_gov	Coefficient	Std. err.	t	P> t	[95% conf. interval]	
fed_aid	23.20462	.5793314	40.05	0.000	22.06858	24.34065
state_aid	29.29737	1.538265	19.05	0.000	26.28092	32.31382
local_taxes	.2887931	.6551799	0.44	0.659	-.9959775	1.573564
calendar_year						
2014	3356002	6389486	0.53	0.599	-9173416	1.59e+07
2015	2297298	6454991	0.36	0.722	-1.04e+07	1.50e+07
2016	2782282	6718582	0.41	0.679	-1.04e+07	1.60e+07
2017	-358776.1	6925177	-0.05	0.959	-1.39e+07	1.32e+07
2018	-1498538	6379024	-0.23	0.814	-1.40e+07	1.10e+07
2019	-3999640	6650776	-0.60	0.548	-1.70e+07	9042153
2020	152241.4	6590537	0.02	0.982	-1.28e+07	1.31e+07
2021	-2617357	6219223	-0.42	0.674	-1.48e+07	9578186
2022	-8499640	6072269	-1.40	0.162	-2.04e+07	3407734
2023	-4637340	6275034	-0.74	0.460	-1.69e+07	7667645
_cons	-3.98e+07	7417739	-5.36	0.000	-5.43e+07	-2.52e+07
sigma_u	73734348					
sigma_e	67757411					
rho	.54216706	(fraction of variance due to u_i)				

With Clustered SE:

gen_gov	Coefficient	Robust std. err.	t	P> t	[95% conf. interval]	
fed_aid	23.20462	3.085513	7.52	0.000	17.14326	29.26598
state_aid	29.29737	15.32198	1.91	0.056	-.8020249	59.39676
local_taxes	.2887931	3.30908	0.09	0.930	-6.211757	6.789343
calendar_year						
2014	3356002	2354863	1.43	0.155	-1270028	7982033
2015	2297298	2397275	0.96	0.338	-2412048	7006644
2016	2782282	3161876	0.88	0.379	-3429092	8993656
2017	-358776.1	4570717	-0.08	0.937	-9337761	8620208
2018	-1498538	6026232	-0.25	0.804	-1.33e+07	1.03e+07
2019	-3999640	7856794	-0.51	0.611	-1.94e+07	1.14e+07
2020	152241.4	7832468	0.02	0.984	-1.52e+07	1.55e+07
2021	-2617357	9441915	-0.28	0.782	-2.12e+07	1.59e+07
2022	-8499640	1.07e+07	-0.79	0.428	-2.96e+07	1.26e+07
2023	-4637340	9904329	-0.47	0.640	-2.41e+07	1.48e+07
_cons	-3.98e+07	3.51e+07	-1.13	0.257	-1.09e+08	2.91e+07
sigma_u	73734348					
sigma_e	67757411					
rho	.54216706	(fraction of variance due to u_i)				

* Note regression coefficients do not change when clustering standard errors, only standard errors (red) and corresponding T statistics (blue)

Clustering Standard Errors

	(1)	(2)	(3)
	gen_gov	gen_gov	gen_gov
fed_aid	14.47*** (30.91)	23.20*** (40.05)	23.20*** (7.52)
state_aid	32.54*** (26.38)	29.30*** (19.05)	29.30 (1.91)
local_taxes	-1.986*** (-16.61)	0.289 (0.44)	0.289 (0.09)
2013.calen~r		0 (.)	0 (.)
2014.calen~r		3356002.0 (0.53)	3356002.0 (1.43)
2015.calen~r		2297298.1 (0.36)	2297298.1 (0.96)
2016.calen~r		2782282.1 (0.41)	2782282.1 (0.88)
2017.calen~r		-358776.1 (-0.05)	-358776.1 (-0.08)
2018.calen~r		-1498538.1 (-0.23)	-1498538.1 (-0.25)

(1) Random Effects

(2) FE (no cluster)

(3) FE (cluster)

2018.calen~r	-1498538.1 (-0.23)	-1498538.1 (-0.25)	
2019.calen~r	-3999640.2 (-0.60)	-3999640.2 (-0.51)	
2020.calen~r	152241.4 (0.02)	152241.4 (0.02)	
2021.calen~r	-2617356.6 (-0.42)	-2617356.6 (-0.28)	
2022.calen~r	-8499639.6 (-1.40)	-8499639.6 (-0.79)	
2023.calen~r	-4637339.9 (-0.74)	-4637339.9 (-0.47)	
_cons	-14448868.0*** (-9.34)	-39788691.4*** (-5.36)	-39788691.4 (-1.13)
N	2967	2967	2967

t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001

* Note the change in statistical significance for local_taxes before clustering standard errors and after (highly significant to insignificant after cluster)

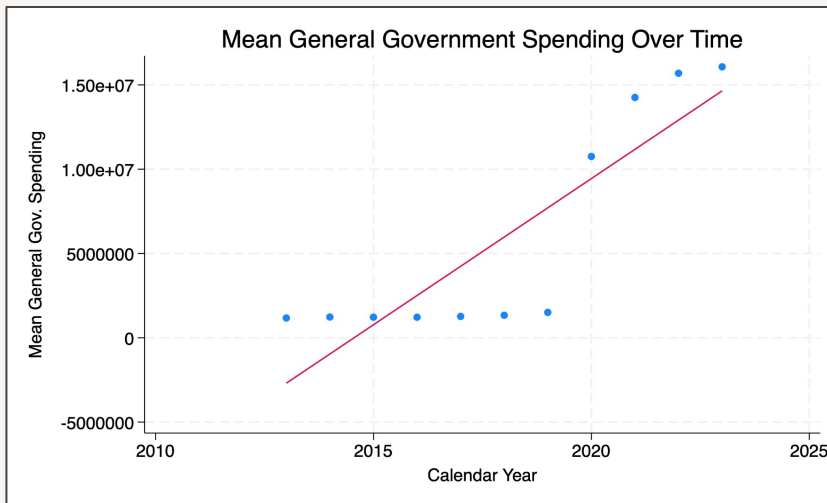
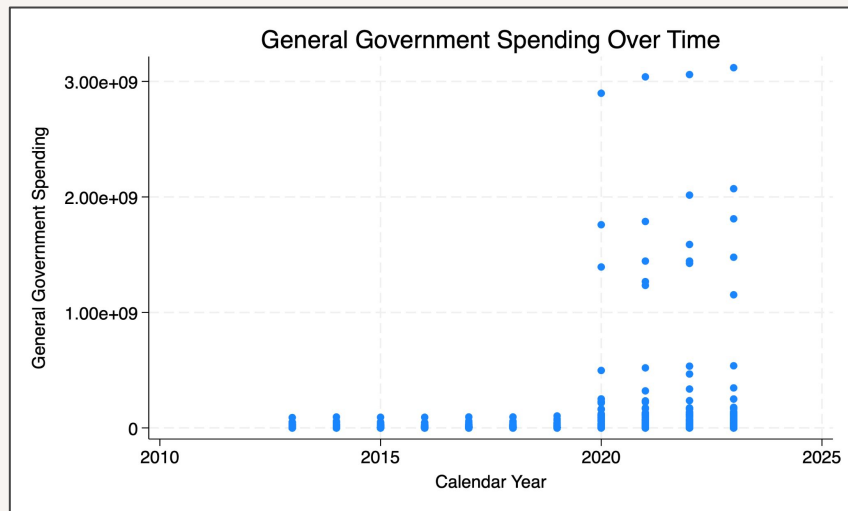
Clustering Standard Errors

Potential Unit Fixed Effects:

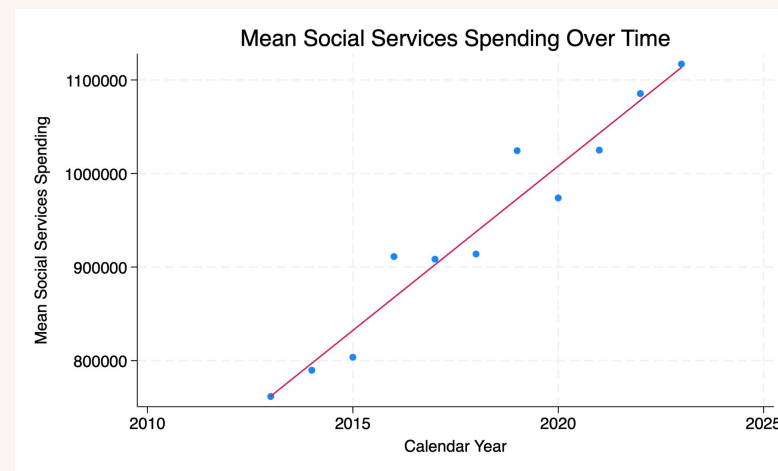
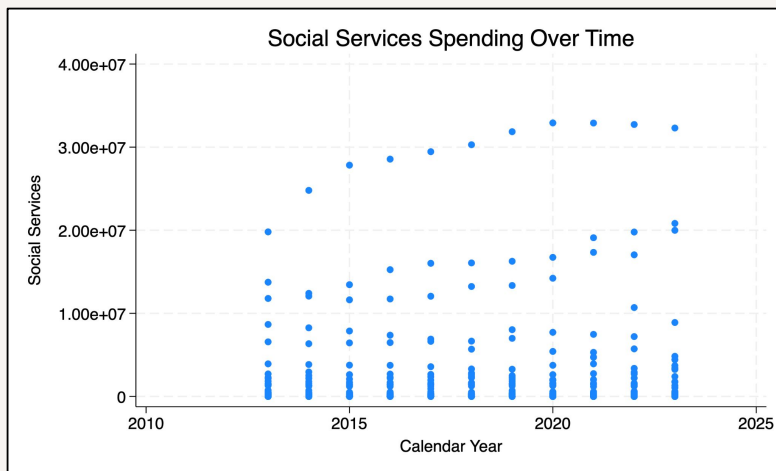
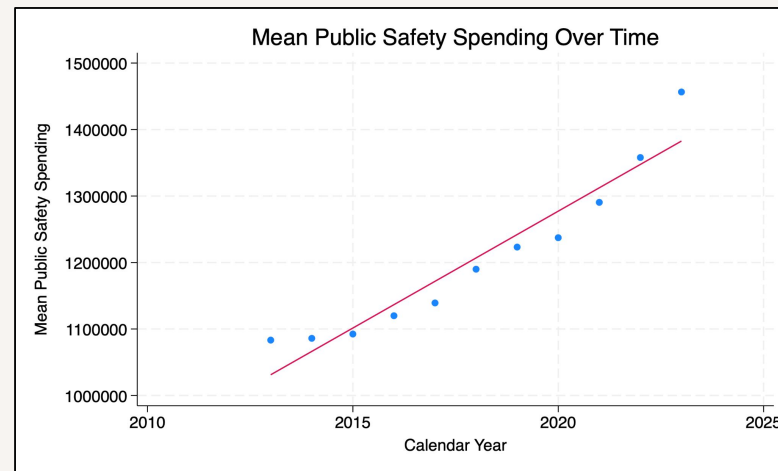
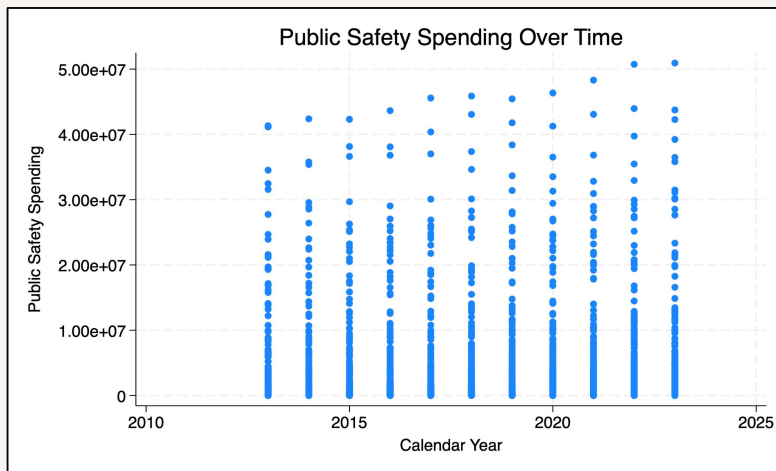
- Sense of community within town
- Urban vs Rural status
- Vacation town status

Potential Time Fixed Effects:

- COVID
- Political Regime changes
- Spending Contributions to Pensions (Large spike in 2019)



* Graphs are shown in dollars, not thousands



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Findings

	(1) gen_gov	(2) public_saf~y	(3) social_ser~s
fed_aid	23.20*** (7.52)	0.0484*** (4.21)	0.00968 (0.37)
state_aid	29.30 (1.91)	0.0338 (0.59)	-0.0946 (-1.46)
local_taxes	0.289 (0.09)	0.174*** (9.52)	0.0387 (0.88)
2013.calen~r	0 (.)	0 (.)	0 (.)
2014.calen~r	3356002.0 (1.43)	-11858.9 (-0.27)	-1362.2 (-0.02)
2015.calen~r	2297298.1 (0.96)	-2281.4 (-0.05)	38444.1 (0.32)
2016.calen~r	2782282.1 (0.88)	20623.1 (0.36)	56390.0 (0.40)
2017.calen~r	-358776.1 (-0.08)	15922.0 (0.33)	57228.0 (0.37)
2018.calen~r	-1498538.1 (-0.25)	45549.4 (1.01)	80978.3 (0.54)
2019.calen~r	-3999640.2 (-0.51)	95038.1 (1.65)	109849.2 (0.68)

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2020.calen~r	152241.4 (0.02)	23381.6 (0.48)	144007.8 (1.01)
2021.calen~r	-2617356.6 (-0.28)	71370.1 (1.43)	204280.8 (1.27)
2022.calen~r	-8499639.6 (-0.79)	97063.8* (2.12)	209250.9 (1.42)
2023.calen~r	-4637339.9 (-0.47)	231862.8*** (4.20)	135966.5 (0.97)
_cons	-39788691.4 (-1.13)	482746.7* (2.19)	117296.8 (0.10)
N	2967	2967	604
t statistics in parentheses * p<0.05, ** p<0.01, *** p<0.001			

* 2013 is reference year for LSDV Time Fixed Effects

Findings

	(1) gen_gov	(2) public_saf~y	(3) social_ser~s
fed_aid	23.20*** (7.52)	0.0484*** (4.21)	0.00968 (0.37)
state_aid	29.30 (1.91)	0.0338 (0.59)	-0.0946 (-1.46)
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2018.calen~r	-1498538.1 (-0.25)	45549.4 (1.01)	80978.3 (0.54)
2019.calen~r	-3999640.2 (-0.51)	95038.1 (1.65)	109849.2 (0.68)

General Government spending:
mainly driven by state and federal
aid

Public Safety spending: mainly
driven by local taxes

Social Services spending: mainly
driven by federal aid, but local
taxes do contribute

Findings

Within towns in NY:

- Most of the general spending comes from government aid
- Local taxes are the main contributor to public safety spending
- Federal aid and local taxes contribute highest to social services spending

Extensions:

- Expand the data outside of just NY, possibly Tri-state area or entire US
- Expand data for more years than just from 2013–2023
- Control for more variables (population, political regime, median income)

Thank you